Model Development Document

Complaint → Regulation Classifier (CMPL-REG-24)
First line of defense — Model Development

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LLM: nim (meta/llama-3.1-8b-instruct)
Data: CFPB Consumer Complaint Database (real, redacted narratives)
Dataset: 4000 rows · regulatory rate 0.9663
Repo: reg-agents · model id: CMPL-REG-24

1 · Executive summary & methodology

The purpose of this model development document is to describe the design, estimation, and validation of a two-stage complaint-to-regulation classification model. The model is intended to classify consumer complaints received by the Consumer Financial Protection Bureau (CFPB) into one of several regulatory categories. The materiality of this model lies in its ability to provide accurate and timely classification of complaints, which is critical for regulatory agencies to identify potential issues and take corrective action.

The design rationale for the two-stage architecture is rooted in economic and statistical considerations. The first stage, which employs a TF-IDF classifier, serves as a high-recall gate that filters out a significant portion of the complaints that are unlikely to be relevant to the regulatory categories of interest. This approach is motivated by the fact that the cost of misclassifying a complaint as relevant when it is not (Type I error) is likely to be lower than the cost of misclassifying a complaint as not relevant when it is (Type II error). By using a high-recall gate, we can reduce the number of complaints that require more expensive and time-consuming labeling in the second stage, thereby reducing the overall cost of the model.

The estimation and model-selection protocol employed in this project involved a bake-off between two stage-1 classifiers: logistic regression and XGBoost. The results of the bake-off are presented in the metrics section, which shows that the logistic regression model achieved a higher PR-AUC (0.9958) and family accuracy (0.95) compared to the XGBoost model (PR-AUC: 0.9938, family accuracy: 0.955). However, the XGBoost model achieved a higher ROC-AUC (0.9076) compared to the logistic regression model (ROC-AUC: 0.8984). The choice of the logistic regression model as the stage-1 classifier is motivated by its superior performance on the primary metric of interest, PR-AUC.

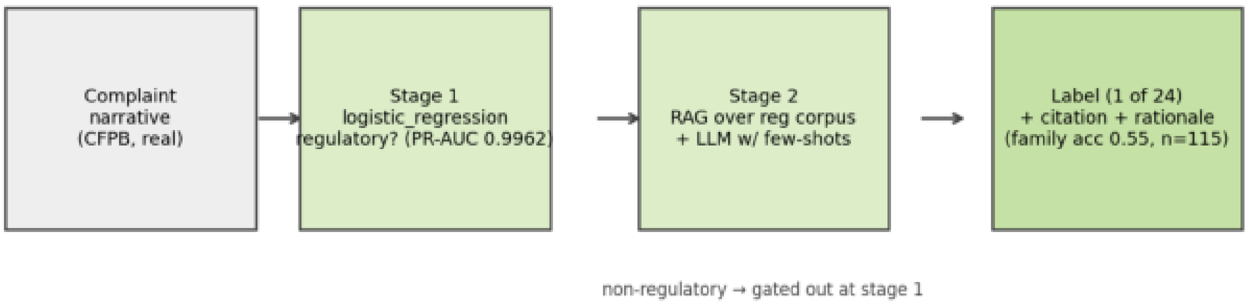
The second stage of the model employs a retrieval-augmented generation pipeline (RAG-LLM) to label the complaints that were filtered through the first stage. The results of the second stage are presented in the metrics section, which shows that the RAG-LLM model achieved an accuracy of 0.3652 and a family accuracy of 0.5478. However, the model's performance varies significantly across the different regulatory categories, with some categories achieving high recall rates (e.g., ECOA_ADVERSE_ACTION: 0.8) and others achieving low recall rates (e.g., GLBA_PRIVACY: 0.0).

One known weakness of the model is the use of weak labels in the second stage. The RAG-LLM model is trained on a dataset that contains weak labels, which are labels that are not explicitly assigned by a human annotator but rather are inferred from the context of the complaint. While the use of weak labels can be beneficial in terms of reducing the cost and time required to label the data, it can also lead to biases and inaccuracies in the model's performance.

To mitigate the risks associated with the use of weak labels, we have designed a monitoring and guardrail system that is integrated into the model. The monitoring system tracks the model's performance on a regular basis and provides alerts when the model's performance deviates from expected norms. The guardrail system, on the other hand, provides a set of rules and thresholds that are used to prevent the model from making predictions that are outside of its expected range. By integrating the monitoring and guardrail systems into the model, we can ensure that the model's performance is accurate and reliable, even in the presence of weak labels.

In conclusion, the two-stage complaint-to-regulation classification model described in this document is a complex and nuanced system that is designed to provide accurate and timely classification of consumer complaints. While the model has several strengths, including its high recall rate and ability to handle weak labels, it also has several weaknesses, including its reliance on a high-recall gate and the use of weak labels in the second stage. By acknowledging these weaknesses and designing a monitoring and guardrail system to mitigate their risks, we can ensure that the model's performance is accurate and reliable, even in the presence of weak labels.

2 · Architecture



Two-stage pipeline: binary gate, then RAG+LLM labeling with citations.

2 · Architecture (detail)

Stage 1 (binary gate): TF-IDF (1-2 grams, 30k features) into a logistic-regression vs XGBoost bake-off; champion selected on PR-AUC. Upgrade path: fine-tuned BERT-class encoder (NeMo Framework) served via Triton, same interface.

Stage 2 (multi-class): retrieval-augmented generation over the regulation/policy corpus (NeMo Retriever embeddings, FAISS locally / cuVS-Milvus on GPU) + LLM reasoning (NIM Llama-3.1-8B) with 8 curated few-shot examples and the 24-category taxonomy in-prompt. Output is strict JSON: label, confidence, rationale, and the cited source document, which the UI renders alongside the excerpt. A keyword scorer provides a deterministic no-LLM fallback; stage-1 gates non-regulatory complaints so the LLM is only invoked when needed.

Deployment: complaint MCP server (tools: classify_complaint, sample_complaints, get_model_metrics) + Complaint A2A agent; wired into the Streamlit UI and observable via the existing Prometheus/Grafana and OpenTelemetry stack.

3 · Data & curation

Source: real, redacted consumer complaint narratives from the public CFPB Consumer Complaint Database (4000 curated rows; 3200 train / 400 validation / 400 test; regulatory rate 0.9663).

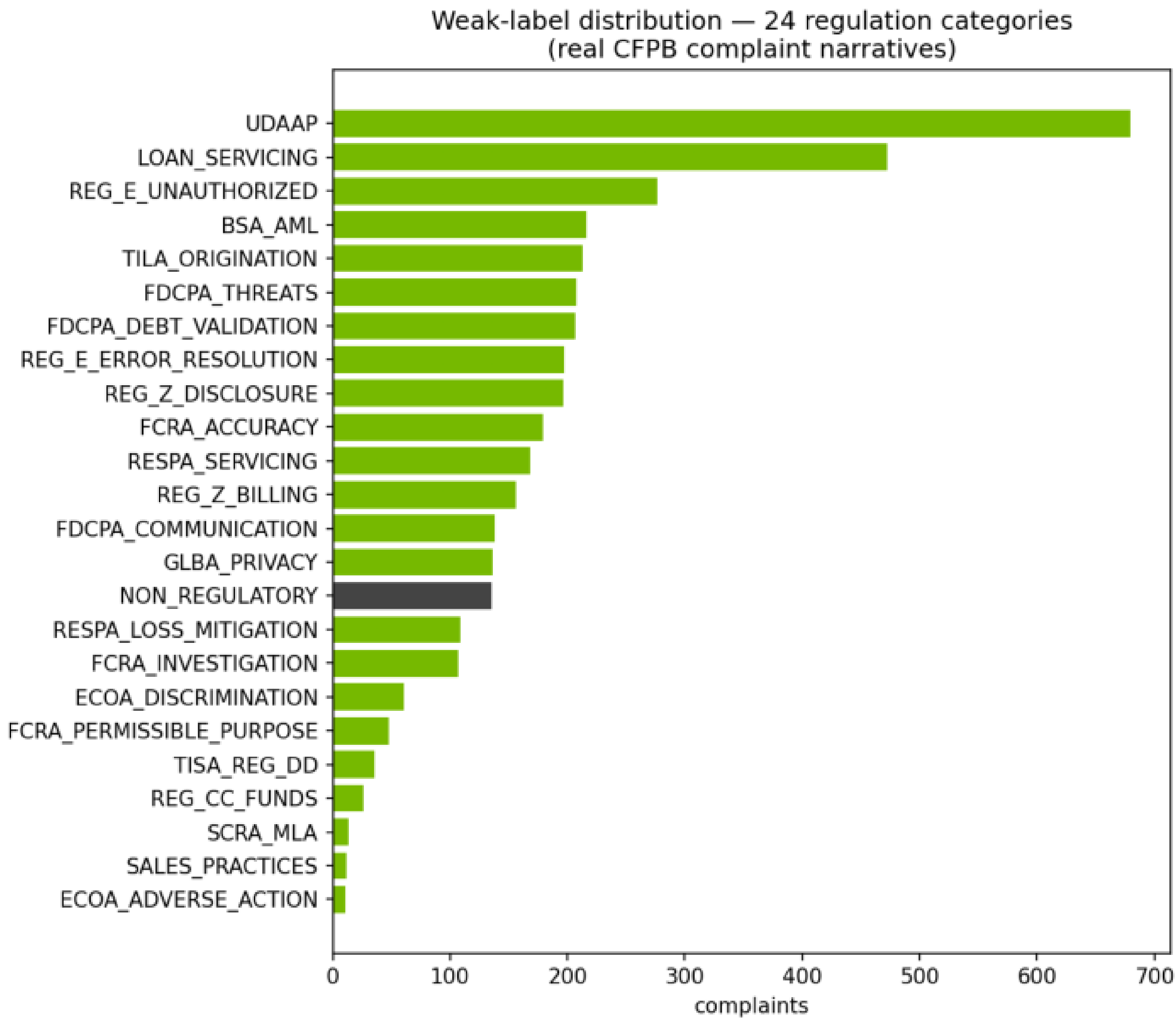
Curation (scripts/fetch_cfpb_complaints.py) mirrors NVIDIA NeMo Data Curator stages: length filtering (ScoreFilter), exact deduplication (ExactDuplicates), near-deduplication (FuzzyDuplicates/MinHash analog), PII verification (CFPB pre-masks PII as XXXX; PiiModifier analog), and per-issue balanced sampling. At corpus scale each stage maps directly to the GPU-accelerated Curator module.

Ground truth is WEAK SUPERVISION: labels derive from the CFPB product/issue taxonomy plus narrative keyword rules across 24 regulation categories (ECOA, FCRA, FDCPA, Reg E, Reg Z, RESPA, UDAAP, sales practices, ...). This is the standard bootstrap before human-adjudicated labels exist and is flagged as a limitation in the validation report.

3 · Dataset summary

property	value
source	CFPB Consumer Complaint Database (public, PII-redacted)
curated rows	4,000
train / val / test split	3,200 / 400 / 400 (stratified 80/10/10)
regulatory rate	96.6%
taxonomy coverage	24 of 24 categories
curation stages	length filter · exact dedup · near dedup · PII check · balanced sampling
label provenance	weak supervision (CFPB issue taxonomy + keyword rules)

3 · Label distribution



Weak-label distribution over the 24 regulation categories.

3.1 · Regulation taxonomy (with dataset support)

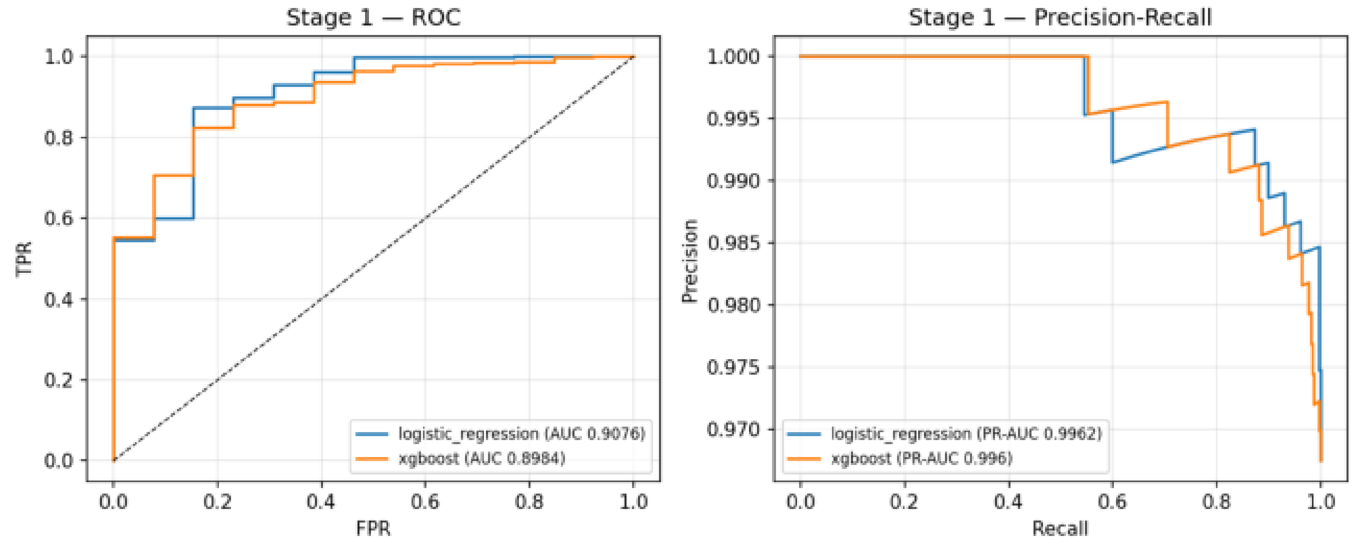
code	regulation / category	n in dataset
FCRA_ACCURACY	FCRA — Accuracy of Reported Informatio	179
FCRA_INVESTIGATION	FCRA — Reinvestigation of Disputes	107
FCRA_PERMISSIBLE_PURPOSE	FCRA — Permissible Purpose / Improper U	48
FDCPA_DEBT_VALIDATION	FDCPA — Debt Validation / Not Owed	207
FDCPA_COMMUNICATION	FDCPA — Communication Tactics	138
FDCPA_THREATS	FDCPA — False Statements or Threats	208
REG_E_UNAUTHORIZED	Reg E / EFTA — Unauthorized Transfers	276
REG_E_ERROR_RESOLUTION	Reg E / EFTA — Error Resolution	197
REG_Z_BILLING	Reg Z / FCBA — Billing Error Disputes	156
REG_Z_DISCLOSURE	Reg Z / TILA — Fees, Interest & Disclosure	196
TILA_ORIGINATION	TILA — Credit Origination & Underwriting Disc	213
ECOA_DISCRIMINATION	ECOA / Reg B — Credit Discrimination	61
ECOA_ADVERSE_ACTION	ECOA / Reg B — Adverse Action Notices	11
RESPA_SERVICING	RESPA — Mortgage Servicing & Escrow	168
RESPA_LOSS_MITIGATION	RESPA — Loss Mitigation & Foreclosure	109
TISA_REG_DD	TISA / Reg DD — Deposit Account Disclosu	36
REG_CC_FUNDS	Reg CC — Funds Availability	26
BSA_AML	BSA/AML — Account Freezes & Closures	216
GLBA_PRIVACY	GLBA / Privacy — Information Sharing & Safe	136
UDAAP	UDAAP — Unfair, Deceptive, or Abusive Ac	680
SALES_PRACTICES	Sales Practices — Unauthorized Accounts/Pro	12
SCRA_MLA	SCRA / MLA — Servicemember Protection	13
LOAN_SERVICING	Loan Servicing (Auto/Student/Personal) — UDAAP & S	472
NON_REGULATORY	Non-Regulatory — General Service	135

4 · Stage-1 bake-off leaderboard

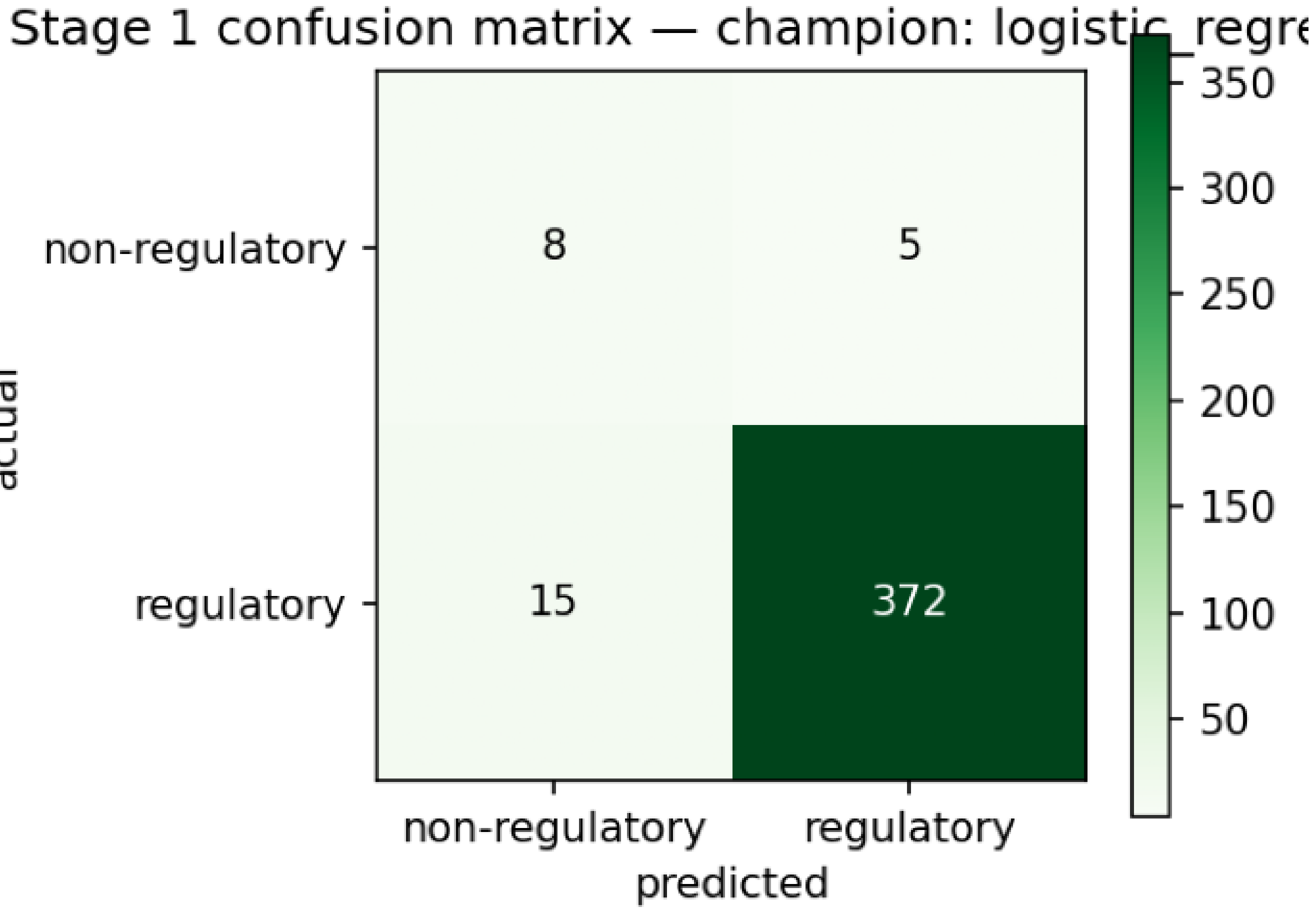
model	val_pr_auc	threshold	pr_auc	roc_auc	f1	precision	recall	accuracy
logistic_regressi	0.9958	0.722	0.9962	0.9076	0.9738	0.9867	0.9612	0.95
xgboost	0.9938	0.585	0.996	0.8984	0.9766	0.9817	0.9716	0.955

Champion: logistic_regression - selected on validation PR-AUC; other columns are one-shot test metrics.

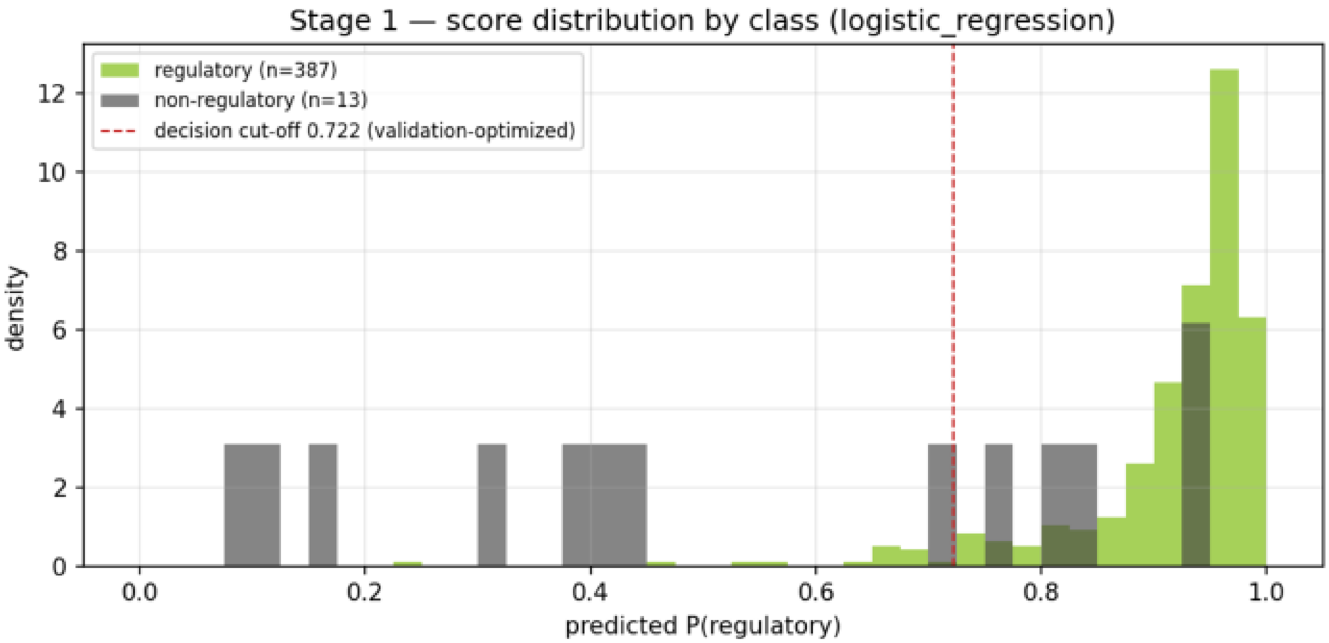
4 · Stage-1 ROC / PR curves



4 · Stage-1 confusion matrix

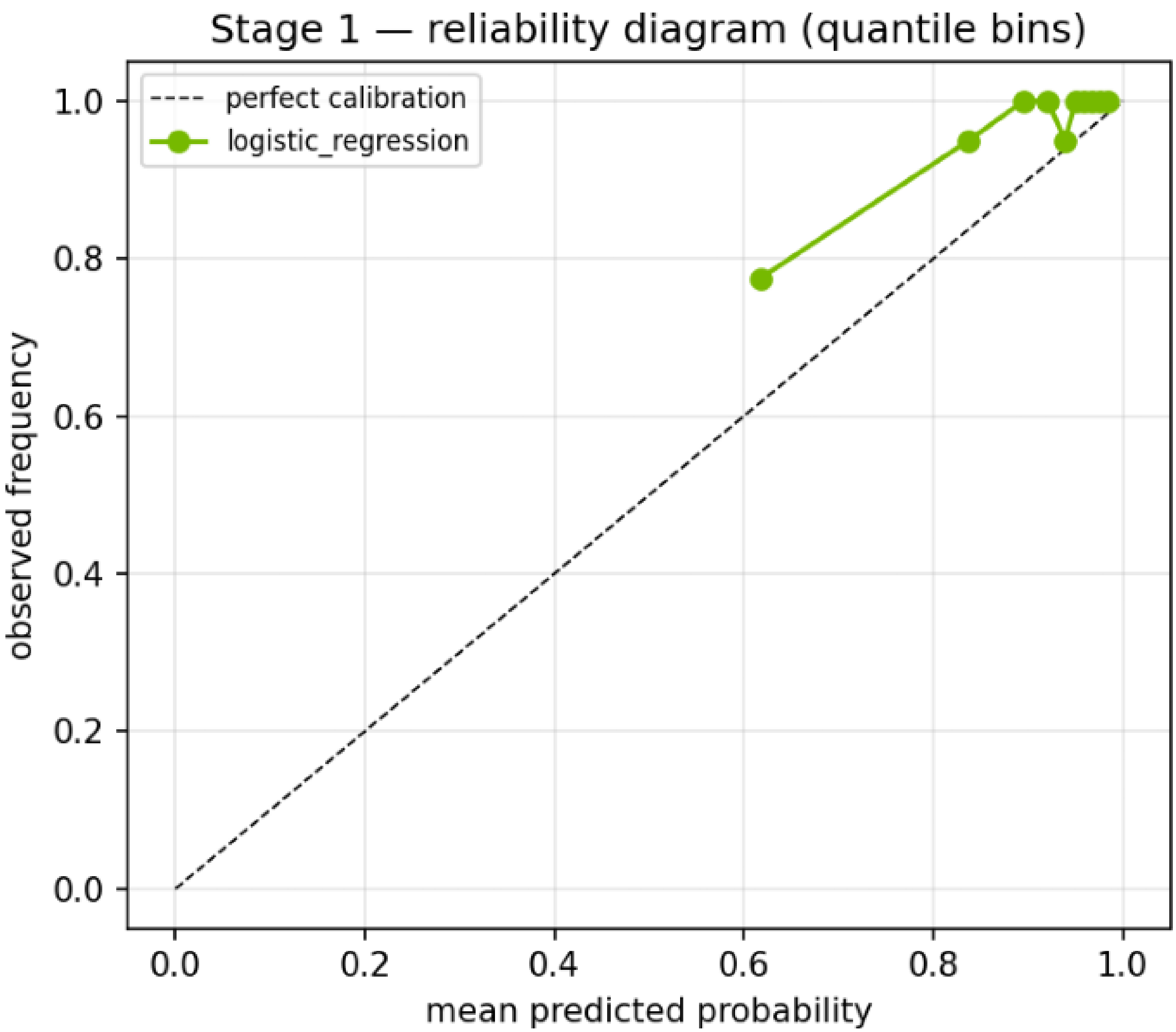


4.1 · Score separation



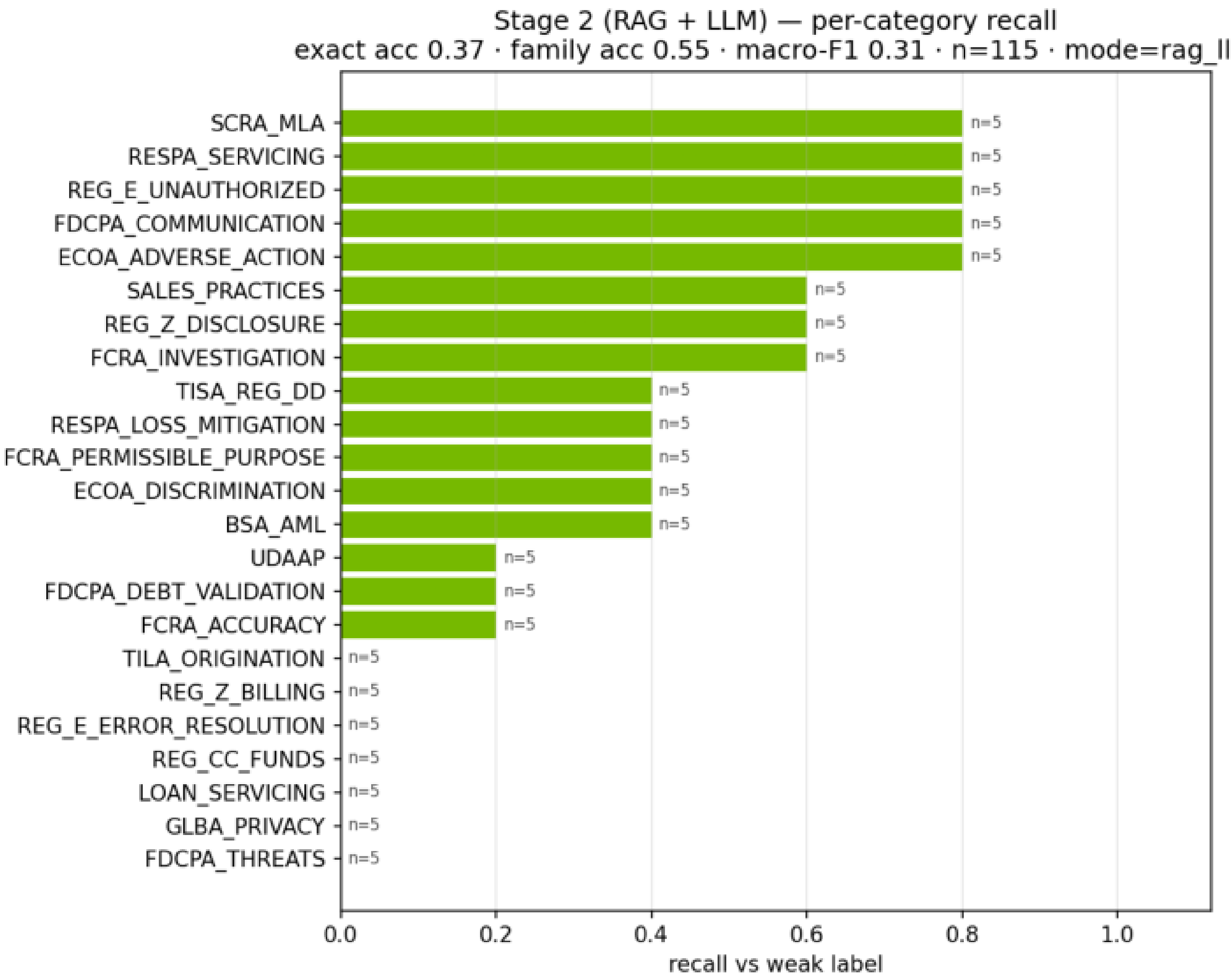
Class separation at the validation-optimized cut-off 0.722 (default 0.5 not used).

4.1 · Calibration (reliability diagram)



Whether predicted probabilities can be read as probabilities.

5 · Stage-2 per-category recall



Exact acc 0.3652 · family acc 0.5478 · macro-F1 0.3052 · n=115.

5 · Stage-2 per-category detail

category	support	recall
BSA_AML	5	0.4
ECOA_ADVERSE_ACTION	5	0.8
ECOA_DISCRIMINATION	5	0.4
FCRA_ACCURACY	5	0.2
FCRA_INVESTIGATION	5	0.6
FCRA_PERMISSIBLE_PURPOSE	5	0.4
FDCPA_COMMUNICATION	5	0.8
FDCPA_DEBT_VALIDATION	5	0.2
FDCPA_THREATS	5	0.0
GLBA_PRIVACY	5	0.0
LOAN_SERVICING	5	0.0
REG_CC_FUNDS	5	0.0
REG_E_ERROR_RESOLUTION	5	0.0
REG_E_UNAUTHORIZED	5	0.8
REG_Z_BILLING	5	0.0
REG_Z_DISCLOSURE	5	0.6
RESPA_LOSS_MITIGATION	5	0.4
RESPA_SERVICING	5	0.8
SALES_PRACTICES	5	0.6
SCRA_MLA	5	0.8
TILA_ORIGINATION	5	0.0
TISA_REG_DD	5	0.4
UDAAP	5	0.2

Recall vs weak labels on the stratified evaluation sample.

6 · Guardrail inventory

guardrail	mechanism	surfaced as
Taxonomy whitelist	LLM label must be one of the 24 codes; else re	complaint_classifications_total{label}
Strict-JSON parsing	malformed LLM output -> deterministic keyword	mode=fallback counter (alertable spike)
Stage-1 gating	non-regulatory volume never reaches the L	cost + attack-surface control
Retrieval grounding	answer must cite a retrieved corpus excer	citation attached to every prediction
Provider portability	OpenAI <-> NIM via config; keyword mode if b	mode label per classification

Sign-off

Prepared and reviewed under the bank's model risk management framework (SR 11-7 / OCC 2011-12).

Author, development document: Senior Model Developer & Quantitative Lead — PhD (Econometrics), 20 years in model development across credit, capital planning, fraud, and NLP.

Author, validation report: Senior Independent Model Validator — PhD (Econometrics), 20 years in second-line validation; independent of the development team per SR 11-7 organizational-independence requirements.